

ASSIGNMENT NO.9: Data Retrieval, Functions, and Joins in MySQL

Questions:

Q1) Create a MySQL database named CollegeDB and create two tables: OnlineEnrollment and OfflineEnrollment.

OnlineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	

OfflineEnrollment Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	NOT NULL
CourseID	VARCHAR(30)	
City	VARCHAR(30)	
Marks	INT	

Insert at least 6 records into each table. Some records should be common in both tables.

Query-

CREATE DATABASE CollegeDB;

USE CollegeDB;

CREATE TABLE Course (

CourseID INT PRIMARY KEY,

CourseName VARCHAR(50) NOT NULL,

Fees DECIMAL(10,2) NOT NULL

);

CREATE TABLE Student (

```
StudentID INT PRIMARY KEY,  
StudentName VARCHAR(50) NOT NULL,  
CourseID INT,  
City VARCHAR(30),  
Marks INT,  
Email VARCHAR(60),  
FOREIGN KEY (CourseID) REFERENCES Course(CourseID)  
);
```

```
INSERT INTO Course  
(CourseID, CourseName, Fees)  
VALUES  
(101, 'BSc IT', 45000.00),  
(102, 'BCA', 40000.00),  
(103, 'BSc CS', 42000.00),  
(104, 'Data Science', 55000.00);
```

```
INSERT INTO Student  
(StudentID, StudentName, CourseID, City, Marks, Email)  
VALUES  
(1, 'Rahul Sharma', 101, 'Mumbai', 78, 'rahul@gmail.com'),  
(2, 'Priya Patil', 102, 'Pune', 85, 'priya@gmail.com'),  
(3, 'Aman Verma', 103, 'Nashik', 67, 'aman@gmail.com'),  
(4, 'Sneha Joshi', 101, 'Mumbai', 91, 'sneha@gmail.com'),  
(5, 'Karan Mehta', 102, 'Thane', 74, 'karan@gmail.com'),  
(6, 'Neha Shah', 101, 'Pune', 88, 'neha@gmail.com'),  
(7, 'Rohan Desai', 103, 'Mumbai', 69, 'rohan@gmail.com'),  
(8, 'Meera Kulkarni', 102, 'Nashik', 82, 'meera@gmail.com'),
```

(9, 'Arjun Nair', 101, 'Thane', 59, 'arjun@gmail.com'),
 (10, 'Divya More', 102, 'Mumbai', 93, 'divya@gmail.com');

2) Write a MySQL query using UNION to display all unique students from both OnlineEnrollment and OfflineEnrollment.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

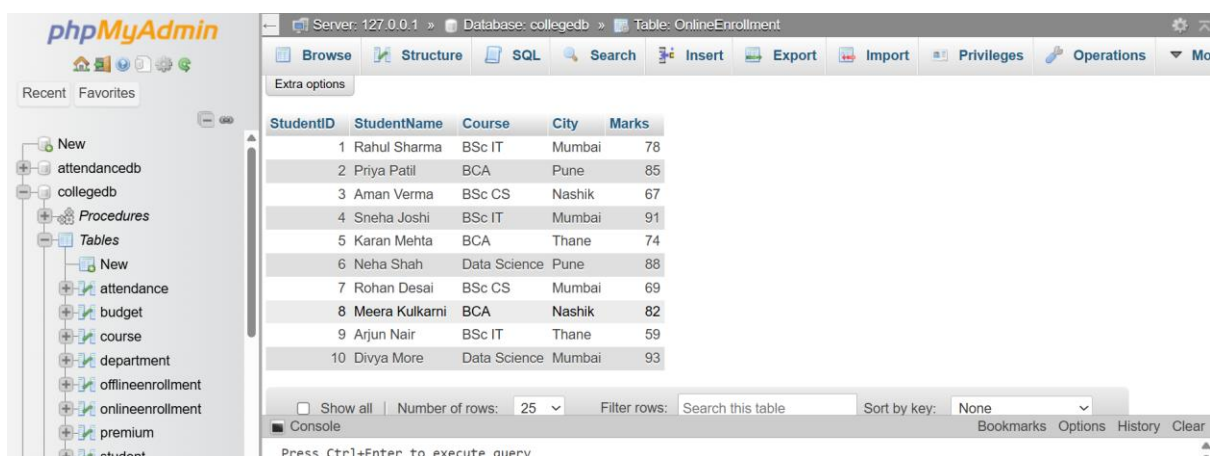
Query-

```
SELECT StudentID, StudentName, Course, City, Marks
FROM OnlineEnrollment
```

UNION

```
SELECT StudentID, StudentName, Course, City, Marks
FROM OfflineEnrollment;
```

Output-



The screenshot shows the phpMyAdmin interface with the 'OnlineEnrollment' table selected. The table contains the following data:

StudentID	StudentName	Course	City	Marks
1	Rahul Sharma	BSc IT	Mumbai	78
2	Priya Patil	BCA	Pune	85
3	Aman Verma	BSc CS	Nashik	67
4	Sneha Joshi	BSc IT	Mumbai	91
5	Karan Mehta	BCA	Thane	74
6	Neha Shah	Data Science	Pune	88
7	Rohan Desai	BSc CS	Mumbai	69
8	Meera Kulkarni	BCA	Nashik	82
9	Arjun Nair	BSc IT	Thane	59
10	Divya More	Data Science	Mumbai	93

Q3) Write a MySQL query using UNION ALL to combine records from both tables while

keeping duplicate records.

Also write one line explaining the difference between UNION and UNION ALL.

Query-

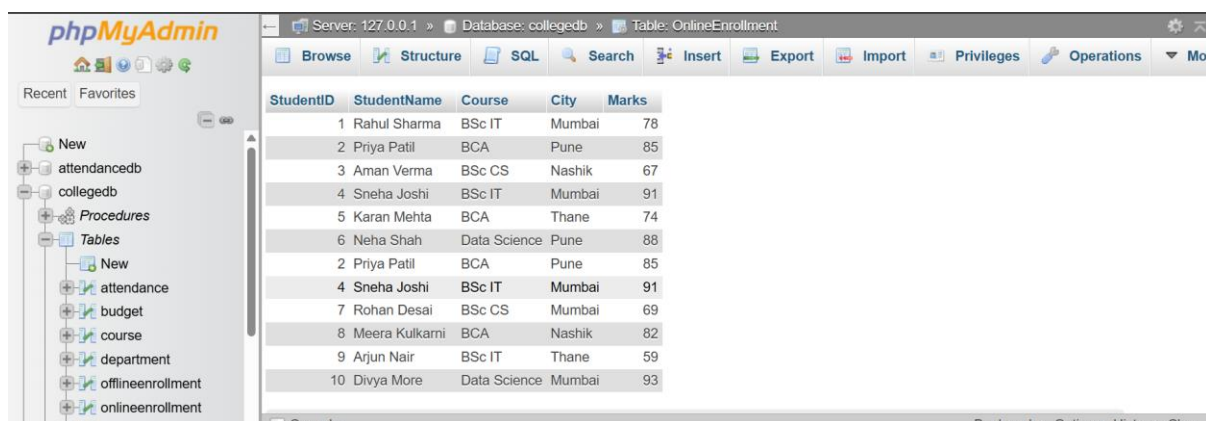
```
SELECT StudentID, StudentName, Course, City, Marks
```

```
FROM OnlineEnrollment
```

UNION ALL

```
SELECT StudentID, StudentName, Course, City, Marks
```

```
FROM OfflineEnrollment;
```



The screenshot shows the phpMyAdmin interface with the 'OnlineEnrollment' table selected. The table contains 10 records. The first 6 records are unique, but records 7 through 10 are duplicates of records 2 through 5 respectively.

StudentID	StudentName	Course	City	Marks
1	Rahul Sharma	BSc IT	Mumbai	78
2	Priya Patil	BCA	Pune	85
3	Aman Verma	BSc CS	Nashik	67
4	Sneha Joshi	BSc IT	Mumbai	91
5	Karan Mehta	BCA	Thane	74
6	Neha Shah	Data Science	Pune	88
2	Priya Patil	BCA	Pune	85
4	Sneha Joshi	BSc IT	Mumbai	91
7	Rohan Desai	BSc CS	Mumbai	69
8	Meera Kulkarni	BCA	Nashik	82
9	Arjun Nair	BSc IT	Thane	59
10	Divya More	Data Science	Mumbai	93

Q4) Write a MySQL query using INTERSECT to display students who are present in both online and offline enrollment tables.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

Query-

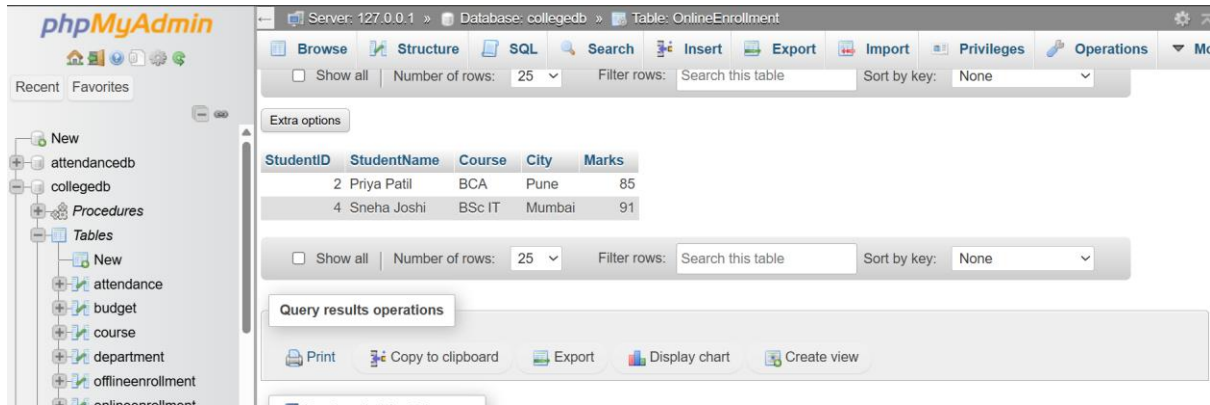
```
SELECT StudentID, StudentName, Course, City, Marks
```

```
FROM OnlineEnrollment
```

INTERSECT

```
SELECT StudentID, StudentName, Course, City, Marks
FROM OfflineEnrollment;
```

Output-



StudentID	StudentName	Course	City	Marks
2	Priya Patil	BCA	Pune	85
4	Sneha Joshi	BSc IT	Mumbai	91

Q5) Write a MySQL query using EXCEPT to display students who are present only in the online enrollment table.

The output should contain:

Column
StudentID
StudentName
Course
City
Marks

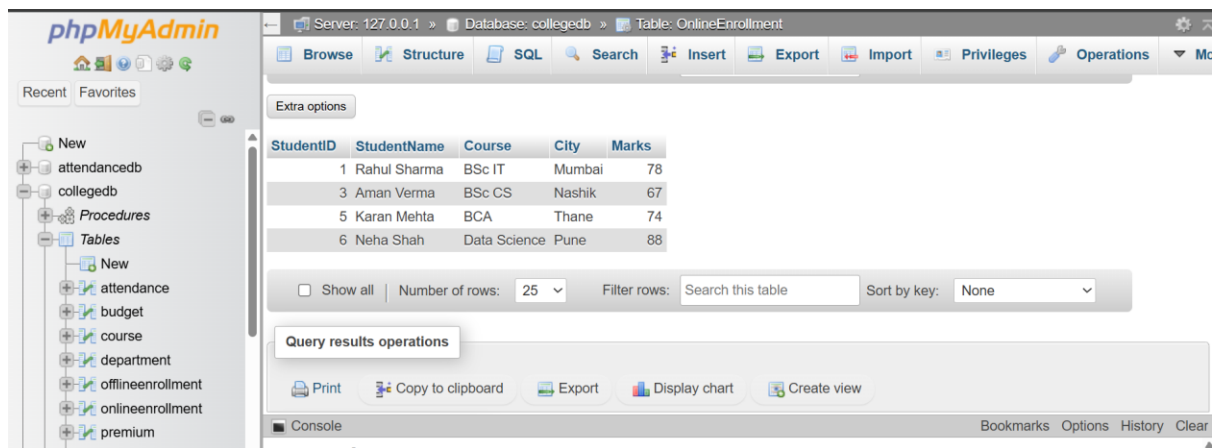
Query-

```
SELECT StudentID, StudentName, Course, City, Marks
FROM OnlineEnrollment
```

```
EXCEPT
```

```
SELECT StudentID, StudentName, Course, City, Marks
FROM OfflineEnrollment;
```

Output-



Q6) Write a MySQL query to display high-scoring students from both tables.
The query should:

Requirement
Select students with marks greater than or equal to 85
Use UNION to combine both tables
Display StudentID, StudentName, Course, and Marks
Sort the final output by Marks in descending order

Query-

```
SELECT StudentID, StudentName, Course, Marks
FROM OnlineEnrollment
WHERE Marks >= 85
UNION
SELECT StudentID, StudentName, Course, Marks
FROM OfflineEnrollment
WHERE Marks >= 85
ORDER BY Marks DESC;
```

Output-

phpMyAdmin

Server: 127.0.0.1 » Database: collegedb » Table: OnlineEnrollment

Browse Structure SQL Search Insert Export Import Privileges Operations M

☐ Profiling [Edit inline] [Edit] [Explain SQL] [Create PHP code] [Refresh]

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Extra options

StudentID	StudentName	Course	Marks
10	Divya More	Data Science	93
4	Sneha Joshi	BSc IT	91
6	Neha Shah	Data Science	88
2	Priya Patil	BCA	85

☐ Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

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